Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

GO!GO!GO!

Project / Solution Name

ChaseLoop

Name (Leader)

PENG YUERAN

Email & Phone (Leader)

University Affiliation(s)

Xiamen University Malaysia

Live Prototype URL

[REPLIT DEPLOYMENT URL]

Code Repository URL (public / access granted to judges)

[GITHUB REPOSITORY URL]

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

ChaseLoop is one FastAPI service on a Replit Deployment with two entry points. The owner dashboard (Jinja2 + HTMX over HTTPS) and the WhatsApp webhook (Meta posts customer replies as JSON) both talk to the same application object and write to PostgreSQL through psycopg.

An in-process APScheduler job wakes once a day, selects invoices due tomorrow and invoices whose last reminder is three days old, asks the LLM API for a message body in the customer's language, and posts it to the WhatsApp Cloud API; every send is written to the message log. When a customer taps one of the three buttons, Meta calls the webhook: the reply is stored, the LLM classifies the intent and extracts any promised date or amount, an attached screenshot is downloaded and read for amount and reference, and a task appears in the owner's queue. The owner verifies the payment and the ledger update is written back. Credentials live in Replit Secrets and never reach the browser. The invoice feed is kept pluggable - form, CSV, or the Orbyte hub in Phase 2 - so no sponsor dependency is needed to run the prototype today.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
FastAPI 0.115 + Uvicorn	HTTP API and ASGI server for the dashboard and webhook	MIT
Jinja2 3.1 + HTMX 2.0	Server-rendered owner dashboard and inline case dialogs	BSD-3-Clause / MIT
APScheduler 3.10	In-process scheduler for the D-1 reminder and the three-day follow-up	MIT
httpx 0.27	Calls to the WhatsApp Cloud API and the LLM API	BSD-3-Clause
psycopg 3.2	PostgreSQL driver for the ledger and message log	LGPL-3.0
pytest 8.3	Test suite for the scheduler, intent rules and webhook handling	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
WhatsApp Cloud API (Meta)	Sends the reminder with three reply buttons, receives the customer reply, downloads payment screenshots	Free tier, then paid per conversation
DeepSeek Chat API (DeepSeek)	Drafts reminder text in English, Bahasa Melayu and Mandarin; classifies reply intent; extracts promised date and amount; reads screenshot fields	Paid (team-funded)
Google Gemini API (gemini-1.5-flash)	Fallback drafting and classification if the primary provider is unavailable	Free tier

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Autoscale)	Hosts the FastAPI app, the scheduler and the WhatsApp webhook	Core plan, Singapore region
Replit PostgreSQL (Neon)	Clients, invoices, messages, reply intents and audit trail	Development tier
Replit Secrets	Runtime credentials: database URL, WhatsApp token, LLM key	Included with the workspace
GitHub	Source control, code review and the submission repository	Free
None	No other cloud service is used in Phase 1	-

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
deepseek-chat (DeepSeek-V3)	DeepSeek (api.deepseek.com)	Pre-trained, called via API; no fine-tuning	Commercial API terms
gemini-1.5-flash	Google AI Studio	Pre-trained, called via API; fallback only	Commercial API terms, free tier
None	-	No self-trained or fine-tuned model is used	-

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic invoice ledger	Generated by seed.py in the repository	Synthetic	MIT (our own)	N
Owner interview notes, 12 sessions	Internal; raw notes not published	Self-collected	Internal use only	N (anonymised)
Synthetic WhatsApp reply samples	Written by the team	Synthetic	MIT (our own)	N
SDC 2026 workshop materials	Provided by the organisers	Private	Organiser terms	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
DeepSeek Chat (2026 model)	Backend logic, test cases, documentation drafting	Moderate
ChatGPT (GPT-5 class)	Debugging help, copy-editing, diagram wording	Minor
GitHub Copilot	Inline code completion in the editor	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI-generated change was read before it was merged. Code from an assistant went through the same review as handwritten code: a diff review by a second member, a test added under tests/, and a full suite run before merge. Message templates were checked by hand against Meta's WhatsApp Business policy. Outputs that touch money - amounts, dates, intent - are never trusted directly: the model's reading is shown to the owner as a suggestion, and the database only changes after a human action. Each member can explain the code they own and reproduce every figure in this document.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☑ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
None	No sponsor-provided resource is used	-

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No official SDC sponsor resource is used in Phase 1, so this does not apply yet. For Phase 2 the LLM provider is deliberately replaceable: drafting and classification sit behind one client module with a provider switch, so moving to a sponsor-provided GenAI API is a configuration change, not a rewrite. Nothing in the prototype depends on sponsor credits or sponsor data to function. If the sponsor API were the only available provider, we would change the base URL and model name and re-run the intent test suite.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

ChaseLoop stores business contact data, not consumer profiles: company and contact name, WhatsApp number, preferred language, invoice number, amount, due date, message history, and the payment screenshots a customer sends. Messages and screenshots are kept for 24 months and then deleted by a scheduled job; the owner can delete a customer record at any time. Data sits in the PostgreSQL instance attached to the Replit deployment. Message text and the customer's reply are sent to the LLM provider for drafting and classification with training disabled, and phone numbers are replaced by a case ID before the call. Keys live in Replit Secrets and never reach the browser.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Three limits matter. First, intent classification can misread a reply - "I will settle it" is not "I have paid" - so the system never closes a case on a model output; only the owner can. Second, screenshot reading is OCR plus a language model and can miss a partly visible amount, so the owner always sees the attachment we used. Third, drafting can produce a tone that is wrong for a long-standing customer, so templates are reviewed per language and the owner can edit the wording before the automatic series starts. Provider outage is absorbed by a retry queue, and reminders fall back to the reviewed static template.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

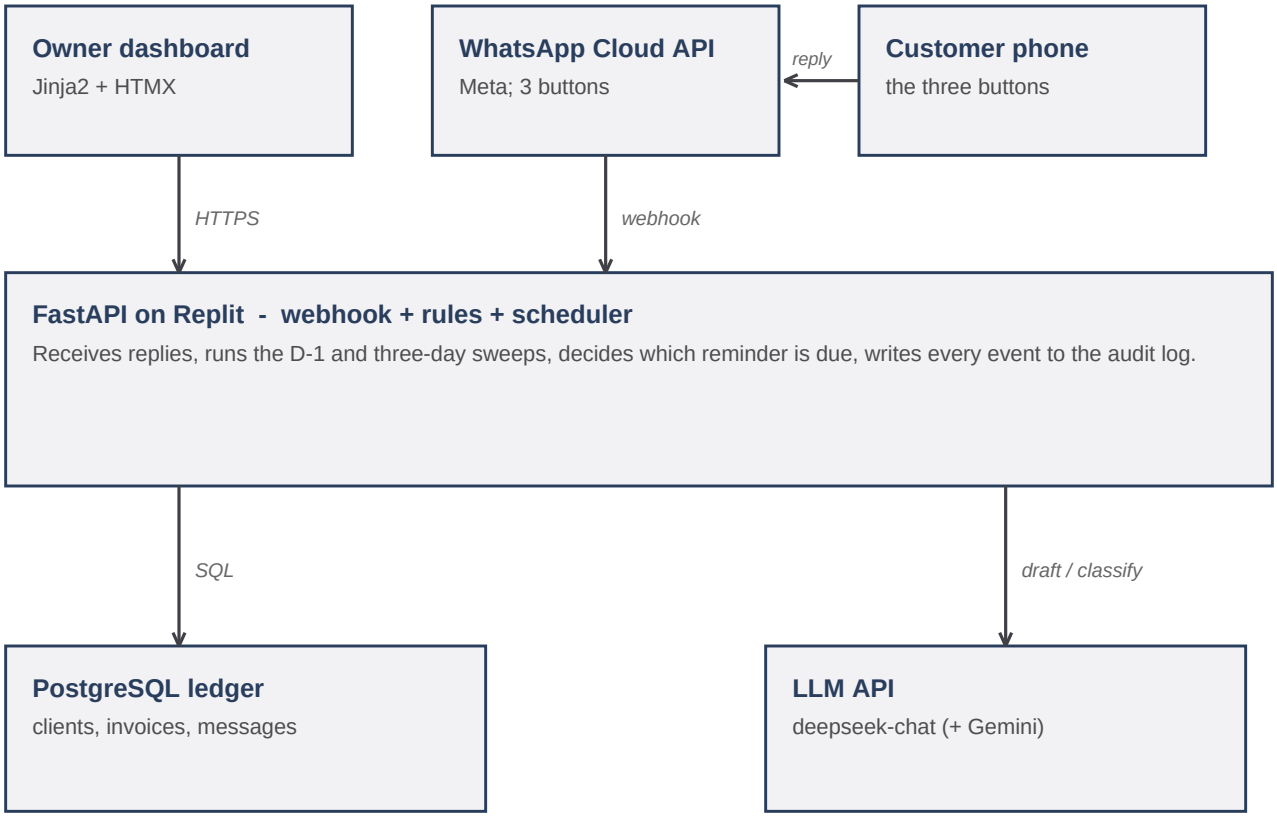
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

PENG YUERAN

24 October 2026

Appendix A - ChaseLoop system block diagram

Solid boxes are components, arrows are data flows. The system sends reminders on its own; every decision that touches money returns to the owner.



Sending is automatic. Money is not.

The scheduler above runs without any approval step: on D-1 and then every three days the reminder leaves the system by itself. Everything a customer sends back lands in the owner's queue - "I have paid" with a screenshot, a request for more time, or a problem - and no case closes until the owner has verified the money. A customer who stays silent for 14 days after the first reminder is escalated to the owner automatically, and both the window and the escalation path are settings the owner controls.